



SECTION:
INSTRUCTOR:

**MIDTERM PART B
SPRING 2026
SOLUTION**

Q 1 (a) A **consistent system of equations** is a system that has **at least one solution**. That solution may be **unique** or there may be **infinitely many solutions**. If a system has **no solution**, it is called **inconsistent**.

(b) Write the augmented matrix:

$$\left[\begin{array}{cc|c} 4 & -6 & b_1 \\ 2 & -3 & b_2 \end{array} \right]$$

Step 1: Swap rows

$$R_1 \leftrightarrow R_2 \Rightarrow \left[\begin{array}{cc|c} 2 & -3 & b_2 \\ 4 & -6 & b_1 \end{array} \right]$$

Step 2: Eliminate the second row

$$R_2 \leftarrow R_2 - 2R_1 \Rightarrow \left[\begin{array}{cc|c} 2 & -3 & b_2 \\ 0 & 0 & b_1 - 2b_2 \end{array} \right]$$

Consistency Condition

For the system to be consistent, the second row must not represent a contradiction.

Thus, the augmented entry must be zero:

$$b_1 - 2b_2 = 0 \Rightarrow b_1 = 2b_2$$

The system is consistent if and only if $b_1 = 2b_2$.

Q 2.

Case 1: Suppose A is not invertible
 $\det(A) = 0$
 Also AB is not invertible
 $\therefore \det(AB) = 0$
 $\det(AB) = \det(A) \det(B)$
 $0 = 0 \det(B) \Rightarrow 0 = 0$ **QED**

THEOREM 1.6.5

Let A and B be square matrices of the same size. If AB is invertible, then A and B must also be invertible.



Q 3 (a)

Let $\mathbf{u} = (u_1, u_2, \dots, u_n)$, $\mathbf{v} = (v_1, v_2, \dots, v_n)$, and $\mathbf{w} = (w_1, w_2, \dots, w_n)$. Then

$$(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n) \cdot (w_1, w_2, \dots, w_n)$$

(b)

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &= (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) \\ &= \mathbf{u} \cdot \mathbf{u} + \mathbf{v} \cdot \mathbf{u} + \mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{v} \\ &= \|\mathbf{u}\|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2 \\ &\leq \|\mathbf{u}\|^2 + 2|\mathbf{u} \cdot \mathbf{v}| + \|\mathbf{v}\|^2 \quad (\text{absolute value prop.}) \\ &\leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 \quad (\text{CS-ineq.}) \\ &\leq (\|\mathbf{u}\| + \|\mathbf{v}\|)^2 \end{aligned}$$



Q 4 (a) A matrix is in **reduced row echelon form (RREF)** if **all** of the following conditions are satisfied:

1. **All nonzero rows are above any rows of all zeros.**
(Any rows that are entirely zero must be at the bottom.)
2. **The first nonzero entry in each nonzero row is 1.**
This entry is called a *leading 1* (or pivot).
3. **Each leading 1 is to the right of the leading 1 in the row above it.**
(Pivots move strictly to the right as you go down the rows.)
4. **Each leading 1 is the only nonzero entry in its column.**
(All other entries in the pivot's column are zero, both above and below.)

(b)

Solution:

Form the augmented matrix:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array} \right].$$

Step 1: $R_3 \leftarrow R_3 - R_1$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 \end{array} \right].$$

Thus the inverse is:

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix}.$$

Step 2: $R_3 \leftarrow R_3 - R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & -1 & 1 \end{array} \right].$$

Q. 5

$$\begin{aligned} & \underline{u} \cdot A \underline{v} \\ &= (A \underline{v})^T \underline{u} & (\underline{u} \cdot \underline{v} &= \underline{v}^T \underline{u}) \\ &= \underline{v}^T (A^T \underline{u}) & (AB)^T &= B^T A^T \text{ \textit{associative law}} \\ &= A^T \underline{u} \cdot (\underline{v}^T)^T & (\underline{u} \cdot \underline{v} &= \underline{v}^T \underline{u}) \\ &= A^T \underline{u} \cdot \underline{v} & (\underline{v}^T)^T &= \underline{v} \end{aligned}$$